

Name

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- The ODE for a tank system is $y' = A y$ with $A = \begin{pmatrix} -\frac{5}{10} & \frac{1}{20} \\ \frac{5}{10} & -\frac{5}{20} \end{pmatrix}$.

Sketch a possible tank system.

Compute the two exponential decay rates of the system

1. The ODE for a tank system is $y' = A y$ with $A = \begin{pmatrix} -\frac{5}{20} & \frac{2}{10} \\ \frac{5}{20} & -\frac{5}{10} \end{pmatrix}$.

Sketch a possible tank system.

Compute the exponential decay rates of the system

- Write down the ODE for the LRC circuit with no voltage source and $R = 10.0$, $C = 0.5$, and $L = 0.5$. Fill in the matrix A that make the ODE the a first order system $\frac{d}{dt} \begin{pmatrix} q \\ i \end{pmatrix} = A \begin{pmatrix} q \\ i \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. What would be a suitable time interval to plot over?

$$A = \begin{pmatrix} & \\ & \end{pmatrix}$$

2. Write down the ODE for the LRC circuit with no voltage source and $R = 100.0$, $C = 0.1$, and $L = 0.1$. Fill in the matrix A that make the ODE the a first order system $\frac{d}{dt} \begin{pmatrix} q \\ i \end{pmatrix} = A \begin{pmatrix} q \\ i \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. What would be a suitable time interval to plot over?

$$A = \begin{pmatrix} & \\ & \end{pmatrix}$$